AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

M251
Object Oriented Programming Using Java

Midterm Examination (MTA)
Spring Semester 2025/2026

Date: 21 / Apr. / 2026

Number of Exam Pages: 3 (including this cover sheet)	Time Allowed: 2 hours

Instructions:

1. This exam consists of three parts. It is required to solve **ALL** of them in th external answer booklet:
   - **Part 1: 5** MCQ (10 marks)
   - **Part 2: 2** short answer questions (20 marks)
   - **Part 3: 3** problem solving questions (30 marks)
2. Go through all questions before you start answering them.
3. It is not allowed to use any external material.
4. It is not allowed to use calculators or any electronic device.
5. At the end of the examination:
   - Check that you have written **your name, student ID and section numb** on the first page of your answer booklet.
   - Place your question paper in the answer booklet and hand over them to th invigilator.

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PART 1: MCQ [10 Marks: 5x2]

Question 1: [10 marks] Read the following questions and choose the best answer.

1. Java is _____.
   a. a strongly typed language b. interpreted c. case sensitive
   d. Only B and C e. All the above

2. What is the output of the following code if `s = "Strong"` ?
   ```
   System.out.println(s.substring(3));
   ```
   a. ong b. rong c. Str
   d. Stro e. o

3. Default constructor _____.
   a. has no arguments b. created by Java c. has **void** as a return type
   d. Only A and B e. All the above

4. Which of the following is **not correct** about ArrayList?
   a. is indexable b. is ordered c. its size is dynamic
   d. prevents duplicate elements e. None of the above

5. In _____ relationship between classes, if the whole (aggregating) object is deleted, all the part (aggregated) objects must be deleted.
   a. Aggregation b. Composition c. Inheritance
   d. Encapsulation e. None of the above

PART 2: Short Answer Questions [20 Marks]

Question 2: [10 marks: 2+2+(3x2)]

Consider the following code and then answer the questions below it.

```
package mta;

public class Student
{
  String name;
}
```

```
package mta;
public class Test {
  public static void main(String[] args) {
    Student s1 = new Student();     // line 1
    System.out.println(s1.name); // line 2
    // code lines
  }
}
```

A. In the class **Test** (line 1), **is it valid** to call the constructor of `Student` although you did not define it in the class `Student`? **Explain**.

B. In the class **Test** (line 2), **is it valid** to access the `name` attribute directly. **Explain**.

C. In the class **Test**, replace the comment "`// code lines`" with code implementing the following requirements:
   1. Declare and create an array of **Student** objects where the length of the array is 2.
   2. Create two **Student** objects and assign them to the two elements of the array.
   3. Using a loop, iterate over all **Student** objects of the array and change their names to **Anas**.

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Question 3: [10 marks: 5x2]

What is the exact output of the following code?

```
import java.util.*;
public class TestArrayList {
  public static void main(String[] args) {
    String[] f = {"Banana", "Orange", "Kiwi", "Apple"};
    ArrayList<String> list = new ArrayList<>(Arrays.asList(f));
    System.out.println(list.get(1));
    list.add(2, "Cherry");        System.out.println(list);
    list.set(1, "Mango");       System.out.println(list);
    Collections.reverse(list);    System.out.println(list);
    System.out.println(list.indexOf("banana"));
  }
}
```

PART 3: Problem Solving Questions [30 Marks]

Question 4: [10 marks: 1+3+4+2]

Write the class **Aircraft** which includes: (follow **OOP principles** you studied in the class)

1. 2 private attributes to store the **weight** of the Aircraft (e.g. **1.5 tonne**) and its **max speed** (e.g. **250 km/hour**).
2. a zero-arg constructor and a multi-arg constructor.
3. getters and setters for the two attributes.
4. **toString()** method which overrides the Object's toString() method.

Question 5: [13 marks: 2+3+1+2+5]

Write the class **Helicopter** that inherits the class **Aircraft**, and includes the following:

1. A private attribute to store the **number of rotors** of the Helicopter (e.g. **2**).
2. A multi-arg constructor which invokes the parent class constructor.
3. getter for the new attribute.
4. **toString()** method which invokes toString() method of the parent class.
5. **equals()** method which overrides the Object's equals() method. All attributes (including the inherited ones) should be compared.

Question 6: [7 marks: 2+2+(3x1)]

Write a Test class to test your classes by performing the following tasks:

1. Declare, create 2 **Helicopter** objects and initialize them by valid values.
2. Print a message explaining if the 2 Helicopters are identical or not.
3. Display on the screen the **number of rotors** of the first Helicopter by calling a suitable method.
4. Change the **weight** of the first Helicopter to **1.6**.
5. Display on the screen all information of the second Helicopter by calling no more than one method.

End of questions

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